



SCEGGS Darlinghurst

2005

**Higher School Certificate
Trial Examination**

Mathematics

This is a TRIAL PAPER only and does not necessarily reflect the content or format of the Higher School Certificate Examination for this subject.

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Diagrams should be drawn in pencil
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1–10
- All questions are of equal value

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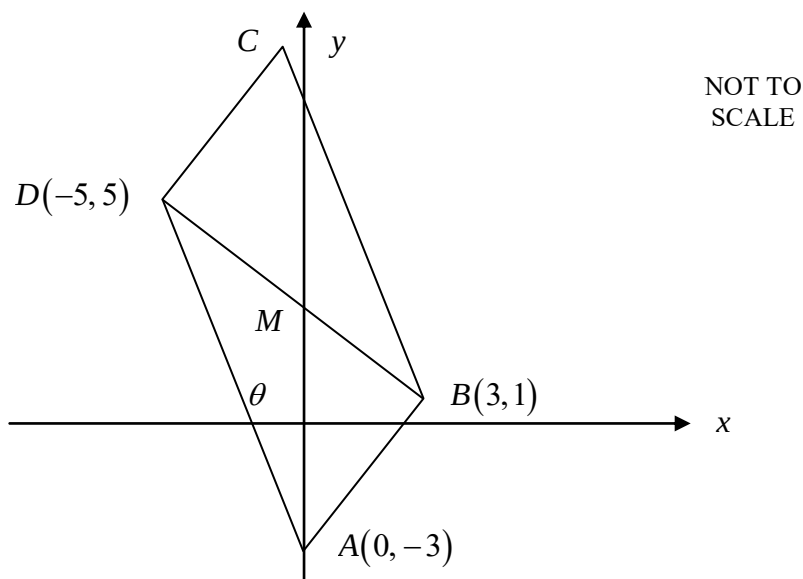
Total marks – 120
Attempt Questions 1–10
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

	Marks
Question 1 (12 marks) Use a SEPARATE writing booklet.	
(a) Find the value of $\frac{1 - 0.46^2}{1 + 0.46^2}$ correct to 3 significant figures.	1
(b) Write $\frac{1}{\sqrt{6} - 2}$ with a rational denominator.	2
(c) If α and β are the roots of $2x^2 - 6x + 3 = 0$, find the values of:	
(i) $\alpha + \beta$	1
(ii) $\alpha\beta$	1
(iii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$	2
(d) Graph the solution of $ 2x + 1 \leq 7$ on a number line.	2
(e) Solve $4^x + 3(2^x) - 28 = 0$ for x .	3

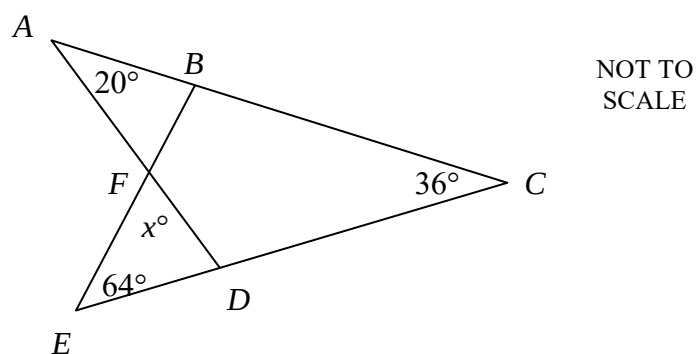
Question 2 (12 marks) Use a SEPARATE writing booklet.

(a)



- (i) Find the co-ordinates of M , the midpoint of BD . 1
- (ii) Find the co-ordinates of C so that $ABCD$ is a parallelogram. 1
- (iii) Show that the line AB has equation $4x - 3y - 9 = 0$. 2
- (iv) Find the perpendicular distance between D and the line AB . 2
- (v) Find the area of parallelogram $ABCD$. 2

(b)



Copy the diagram above into your answer booklet. Find the value of x giving reasons.

- (c) Find the equation of the tangent to the curve $y = \cos 2x$ at $(\pi, 1)$. 2

Question 3 (12 marks) Use a SEPARATE writing booklet.

(a) Differentiate:

(i) $y = 7\sqrt{x} - \frac{2}{x^3}$ 2

(ii) $f(x) = \frac{2x-1}{3x+1}$ 2

(b) Beryl does the following solution to solve the equation $2\cos\theta = 1$ for $-\pi \leq \theta \leq \pi$.

Line 1 $2\cos\theta = 1$

Line 2 $\cos\theta = \frac{1}{2}$

Line 3 $\theta = 60^\circ$

(i) Beryl has made at least one mistake in her working. State the line(s) in which the mistake(s) occurred and describe the mistake(s). 2

(ii) Show the correct solution to the equation. 2

(c) On a visit to Sydney Harbour, Mary and Frederik sail on their yacht, the Dannebrog, from point A on a course of 077° for 20 nautical miles to point B. They then change course to 130° and continue sailing for 30 nautical miles to point C.

(i) Draw a neat sketch (at least $\frac{1}{3}$ page) depicting this information. 1

(ii) Show that $\angle ABC = 127^\circ$ 1

(iii) Given that point A and point C are 45 nautical miles apart find the bearing of point C from their starting point. (Answer correct to the nearest degree.) 2

Question 4 (12 marks) Use a SEPARATE writing booklet.

(a) Given the function:

$$f(x) = \begin{cases} x - 5 & \text{for } x \leq 5 \\ (x - 5)^2 & \text{for } x > 5 \end{cases}$$

Find:

(i) $f(-2)$ 1

(ii) $f(a + 5)$ when $a > 0$ 1

(b) Solve for x 3

$$2 \log_e x = \log_e (6 - 5x)$$

(c) A plant is observed over a period of time. Its initial height is 20 cm. It grows 5 cm during the first week of observation. In each succeeding week the growth, in height, is 80% of the previous week's growth. Assuming this pattern continues, calculate the plant's ultimate height. 3

(d) A certain parabola has a focus of $(3, 6)$ and a directrix $y = 2$.

(i) Draw a diagram showing this information and the approximate position of the parabola. 1

(ii) State the co-ordinates of the vertex. 1

(iii) Write the equation of the parabola in the form $(x - h)^2 = 4a(y - k)$. 1

(e) "Mrs Brimfield is having twins. She could have 2 boys, 2 girls or a boy and a girl. Therefore, the probability that she has 2 boys is $\frac{1}{3}$." 1

Is this statement true or false? Give a reason for your answer.

Question 5 (12 marks) Use a SEPARATE writing booklet.

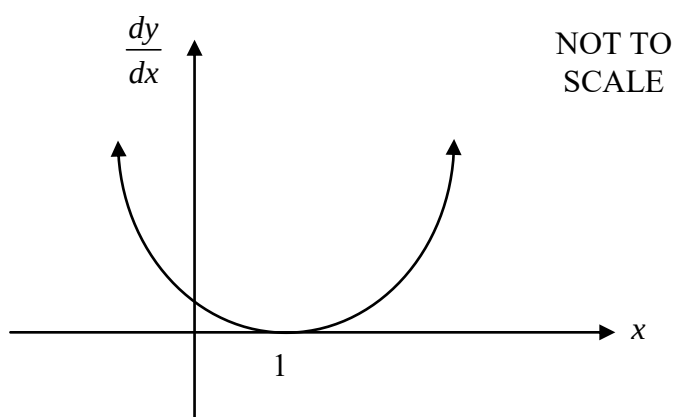
- (a) Find the sum of 10 terms of the series

3

$$\log_m 3 + \log_m 6 + \log_m 12 + \dots$$

given that $\log_m 3 = 0.48$ and $\log_m 2 = 0.30$

- (b) Consider the graph of the derivative $\frac{dy}{dx}$ given below.



- (i) Comment on the sign of $\frac{dy}{dx}$ for all x except $x = 1$.

2

What does this imply about the curve $y = f(x)$ for all x , except $x = 1$?

- (ii) What can you conclude about $y = f(x)$ when $x = 1$?

1

- (iii) Sketch a possible graph of $y = f(x)$

1

Question 5 continues on page 7

Question 5 (continued)

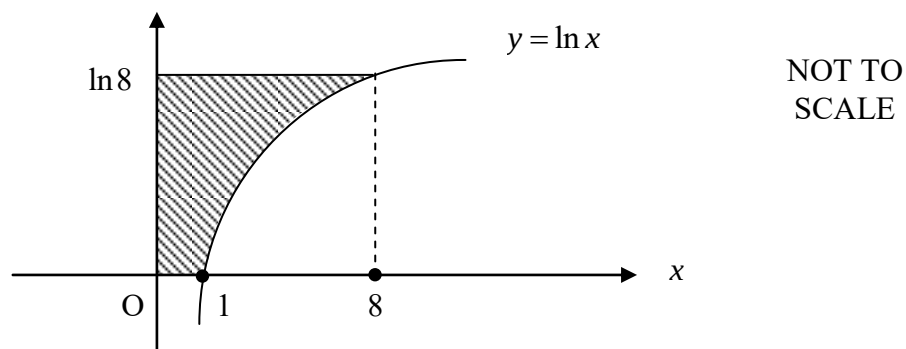
- (c) A census was taken in 2005 of the population of a coalmining town called Blackrock. The population, P , after t years is given by the exponential equation

$$P = 50000 e^{-0.08t}$$

- | | | |
|-------|--|----------|
| (i) | What is the initial population of Blackrock in 2005? | 1 |
| (ii) | Find the time in years it will take the initial population to halve. | 2 |
| (iii) | At what rate is the population changing in 2010? | 2 |

Question 6 (12 marks) Use a SEPARATE writing booklet.

(a)



The diagram shows the area bounded by the graph $y = \ln x$, the co-ordinate axes and the line $y = \ln 8$.

(i) Find the shaded area. 3

(ii) Hence find the exact value at 1

$$\int_1^8 \ln x \, dx$$

(b) (i) Solve $(k-1)(k-9) < 0$ 1

(ii) Find the value of k for which 2

$$kx^2 + (k+3)x + 4$$

is positive definite.

(iii) Explain why $kx^2 + (k+3)x + 4$ is never negative definite. 1

Question 6 continues on page 9

Question 6 (continued)

- (c) (i) Explain why $\int_{-\pi}^{\pi} \sin x \, dx = 0$. **2**
- (ii) Let m be a positive number. With the aid of a clear diagram, find the number of possible solutions for x , so that $\sin x + mx = 0$ in the domain $-\pi \leq x \leq \pi$. **2**

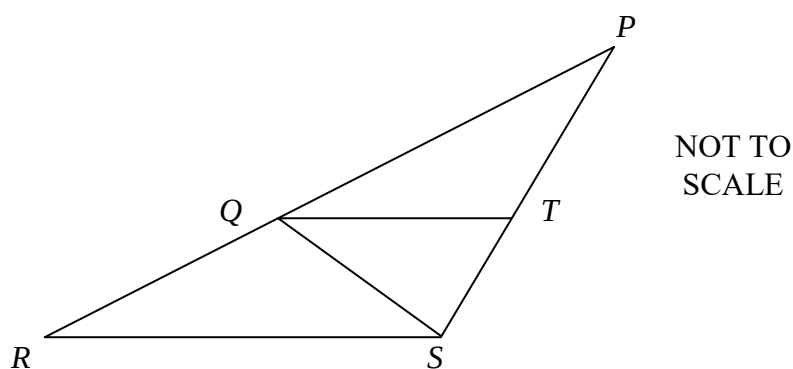
Question 7 (12 marks) Use a SEPARATE writing booklet.

- (a) The sum of the first three terms of a geometric series is 19 and the sum to infinity is 27.

Find:

- | | |
|------------------------------------|---|
| (i) the value of the common ratio. | 2 |
| (ii) the value of the first term. | 1 |
| (iii) the value of the fifth term. | 1 |

- (b)



In the diagram $QT \parallel RS$ and TQ bisects $\angle PQS$.

Copy the diagram into your answer booklet, showing this information.

- | | |
|---|---|
| (i) Explain why $\angle TQS = \angle QSR$. | 1 |
| (ii) Prove that $\triangle QRS$ is isosceles. | 2 |
| (iii) Hence show that $PT : TS = PQ : QS$. | 2 |

Question 7 continues on page 11

Question 7 (continued)

- (c) In a certain hospitality course all students sit for a theory examination in which 60% of the candidates pass.

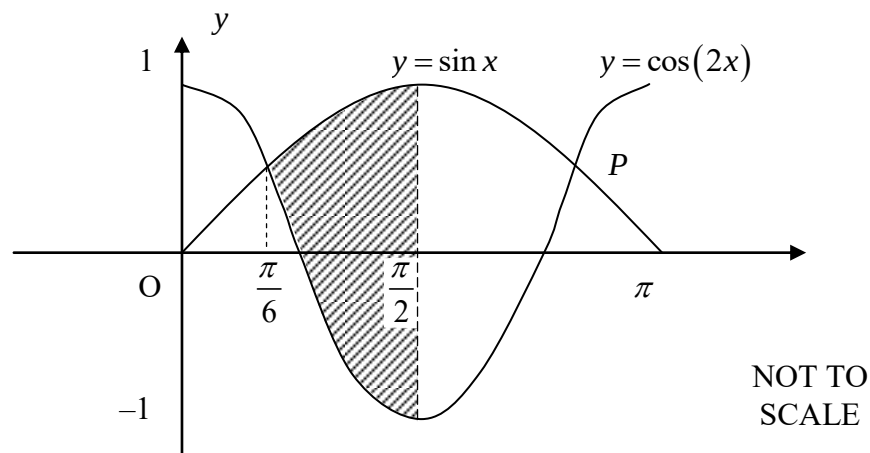
Those who pass the theory examination then sit a practical test which is passed by 40% of those who sit the practical test. A student is chosen at random.

Find the probability that:

- | | | |
|------|--|----------|
| (i) | the student passes both examinations. | 2 |
| (ii) | the student passes just one of the examinations. | 1 |

Question 8 (12 marks) Use a SEPARATE writing booklet.

(a)



The diagram shows parts of the curves $y = \sin x$ and $y = \cos(2x)$.

- (i) The curves intersect at $x = \frac{\pi}{6}$. State the co-ordinates of the point P , the other point of intersection in the domain $0 \leq x \leq \pi$. 2

- (ii) Find the shaded area, leaving your answer in exact form. 4

(b) Consider the function $f(x) = \frac{1}{2}(e^x + e^{-x})$

- (i) Show that the curve represents an even function. 1
- (ii) Show that the function only has one stationary point and determine its nature. 3
- (iii) Show that the function has no points of inflexion. 1
- (iv) Hence sketch the curve. 1

Question 9 (12 marks) Use a SEPARATE writing booklet.

- (a) Heather invests \$50000 in an account that earns 8% p.a. interest, compounded annually. She intends to withdraw \$ M at the end of each year, immediately after the interest has been paid. She wishes to be able to do this for exactly 20 years, so that the account will then be empty.
- (i) Write an expression for the amount of money Heather has in the account immediately after she has made her first withdrawal. 1
- (ii) Write an expression in terms of M for the amount of money in the account, immediately after her 20th withdrawal. 1
- (iii) Calculate the value of M which leaves her account empty after the 20th withdrawal. 2

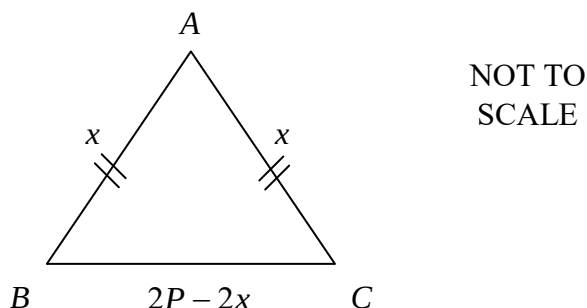
- (b) (i) Copy and complete the table below, correct to 3 decimal places. 2

x	2	3	4	5
$\ln 2x$				

- (ii) Use the table and the Trapezoidal rule to find an approximation for $\int_2^5 \ln 2x \, dx$ correct to 2 decimal places. 1
- (iii) Sketch a graph of $y = \ln 2x$ and use it to explain whether your approximation in (ii) is an over or under estimate of the exact value of the integral. 2
- (iv) Show that $\frac{d}{dx}(x \ln 2x - x) = \ln 2x$ 1
- (v) Hence, deduce the exact value of $\int_2^5 \ln 2x \, dx$. 2

Question 10 (12 marks) Use a SEPARATE writing booklet.

- (a) A machine produces Mathomats of which 5% are defective.
- (i) What is the probability that a Mathomat is NOT defective? 1
- (ii) A random sample of n items is taken from the machine. 2
Find the largest value of n that must be sampled so that the probability that none of the Mathomats are defective is at least 0.5.
- (b) Find the volume of the solid of revolution formed when the area bounded by 3
the curve $y = \frac{1}{\sqrt{2x+1}}$, $x = 0$, $x = 1$ and the x axis is rotated about the x axis.
- (c) $\triangle ABC$ is an isosceles triangle of constant perimeter $2P$ and equal sides of length x .



- (i) Show that the area of the triangle, A , can be given by the expression: 2

$$A = (P - x) \sqrt{2Px - P^2}$$

- (ii) Show that $\frac{dA}{dx} = \frac{P(P - x)}{\sqrt{2Px - P^2}} - \sqrt{2Px - P^2}$ 2

- (iii) Hence, show that the maximum area of all isosceles triangles of constant perimeter $2P$ occurs when $\triangle ABC$ is equilateral. 2

End of Paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

1 a) $0.650709805 \div 0.651$ ①

b) $\frac{1}{\sqrt{6}-2} \times \frac{\sqrt{6}+2}{\sqrt{6}+2} = \frac{\sqrt{6}+2}{6-4} = \frac{\sqrt{6}+2}{2}$ ①

Can't cancel 2 and -4.

c) i) $\alpha + \beta = \frac{6}{2} = 3$ ①

ii) $\alpha\beta = \frac{3}{2}$ ①

iii) $\frac{\beta^2 + \alpha^2}{\alpha^2 \beta^2} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$ ①

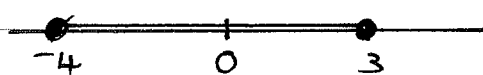
you must learn formulae

$= \frac{3^2 - 2 \times \frac{3}{2}}{(\frac{3}{2})^2} = \frac{9 - 3}{\frac{9}{4}} = \frac{6 \times 4}{9}$
① $= \frac{8}{3} = 2\frac{2}{3}$

Don't guess.

d) $|2x+1| \leq 7$
 $-7 \leq 2x+1 \leq 7$
 $-8 \leq 2x \leq 6$
 $-4 \leq x \leq 3$ ①

Many forget how to do these questions



1 comm

e) Let $m = 2^x$

$m^2 + 3m - 28 = 0$ ①

$(m+7)(m-4) = 0$ ①

$m = -7, 4$

$\therefore 2^x = -7 \quad 2^x = 4$ ①

$\therefore$ no soln + $x = 2$ ①

Reduce to a quadratic equation and you must state $x^2 = -7$ has no solution

must have both

2 a) i) $M = (-1, 3)$ ①

ii) $\frac{x+0}{2} = -1 \quad \frac{y-3}{2} = 3$

$x = -2$

$y - 3 = 6$

$y = 9$ ①

$\therefore C = (-2, 9)$

iii) $m = \frac{4}{3}$ ①

$y - 1 = \frac{4}{3}(x - 3)$ ①

$3y - 3 = 4x - 12$
 $4x - 3y - 9 = 0$

Comm 2

$$(iv) d = \left| \frac{4x - 5 - 3 \times 5 - 9}{\sqrt{16 + 9}} \right|$$

$$= \frac{44}{5}$$

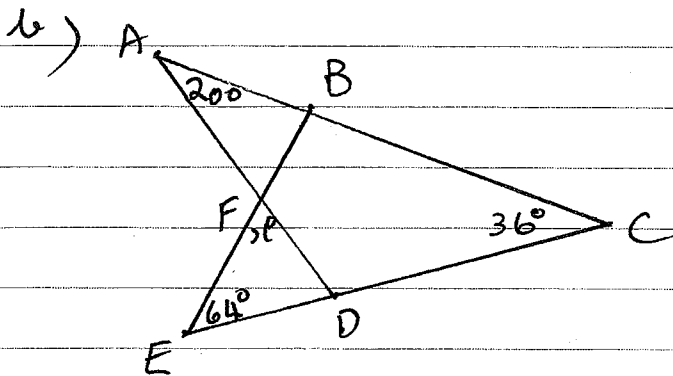
$$(v) AB = \sqrt{(3-0)^2 + (1+3)^2}$$

$$= \sqrt{9+16} = 5$$

$$\therefore \text{Area } ABCD = \frac{44}{5} \times 5$$

$$= 44 \text{ u}^2$$

If using $A = \frac{1}{2} h \times b$ the base is AB, many students used AD.



$$\angle ADC = 180^\circ - 20^\circ - 36^\circ$$

$$= 124^\circ (\angle \text{sum of } \Delta = 180^\circ)$$

$$x^\circ = 124^\circ - 64^\circ = 60^\circ (\text{ext. } \angle \text{ of } \Delta = \text{sum of int. opp } \angle \text{'s})$$

$$c) \frac{dy}{dx} = -2 \sin 2x$$

$$\text{when } x = \pi, m_{\text{tan}} = -2 \sin 2\pi$$

$$= 0$$

$$\therefore y - 1 = 0 (x - \pi)$$

$$\therefore y = 1$$

You must give clear and accurate reasons for every step in your working

Don't forget the negative

evaluate $-2 \sin 2\pi$ to get 0.

$$3) a) (i) y = 7x^{\frac{1}{2}} - 2x^{-3}$$

$$\frac{dy}{dx} = 7 \cdot \frac{1}{2} x^{-\frac{1}{2}} + 6x^{-4}$$

$$= \frac{7}{2\sqrt{x}} + \frac{6}{x^4}$$

learn and practise index laws.

$$(ii) f'(x) = \frac{2(3x+1) - 3(2x-1)}{(3x+1)^2}$$

Calc
4

$$= \frac{6x+2-6x+3}{(3x+1)^2} = \frac{5}{(3x+1)^2}$$

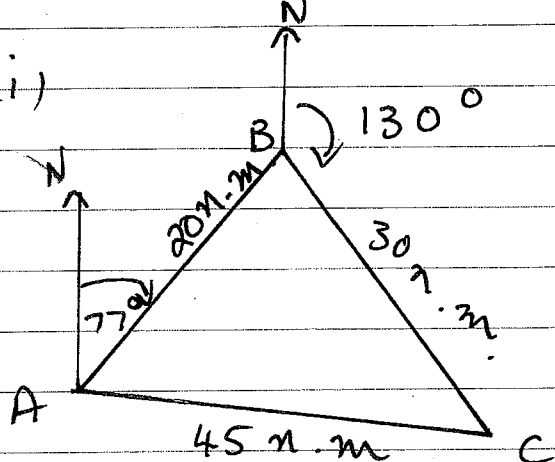
4) i) Line 3. θ must be in radians.

Be brief with your explanations

Line 3. There are 2 solutions

(ii) $\theta = -\frac{\pi}{3}, \frac{\pi}{3}$

c) (i)



1 correct

(ii) $\angle ABN = 180^\circ - 77^\circ = 103^\circ$
(Co-int. $\angle$'s on 11° lines add to 180°)

$\therefore \angle ABC = 360^\circ - 130^\circ - 103^\circ$
 $= 127^\circ$ ($\angle$'s at pt. add to 360°)

(iii) $\cos \angle BAC = (20^2 + 45^2 - 30^2) \div 2 \times 20 \times 45$

$\therefore \angle BAC = 32^\circ$

$\therefore$ The bearing of C from A is 109° T

You can also use the sine rule to find $\angle BAC$. Many did not then state the bearing of C from A.

4) a) (i) $4(-2) = -2 - 5 = -7$

well done

(ii) $4(a+5) = (a+5-5)^2 = a^2$

Simplify in brackets first

b) $\ln x^2 = \ln(6-5x)$

Know log rules.

$\therefore x^2 = 6-5x$

$x^2 + 5x - 6 = 0$

Reas
3

$$(x-1)(x+6)=0 \quad (1)$$

$$\therefore x=1, -6$$

$\therefore x=-6$ is not a soln.

$$\therefore x=1 \quad (1)$$

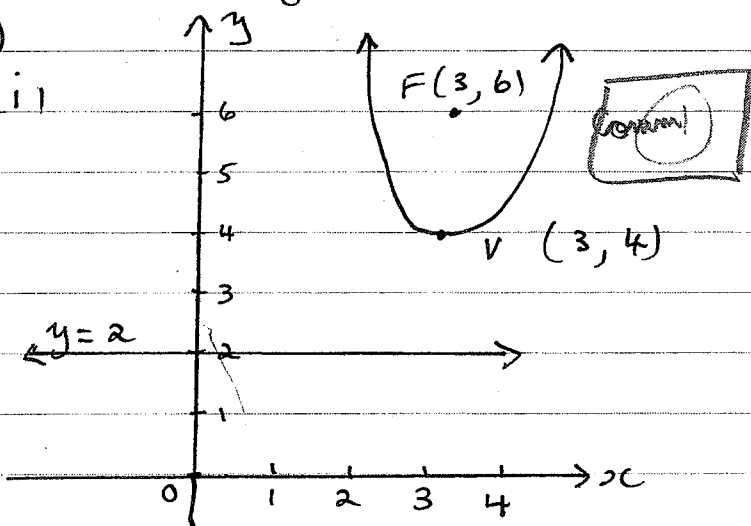
$$c) 20 + 5 + 5 \times 0.8 + 5 \times 0.8^2 + \dots$$

$$S_{\infty} = \frac{5}{1-0.8} = 25 \quad (1)$$

$$\therefore \text{Total height} = 20 + 25 = 45 \text{ cm} \quad (1)$$

d)

i)



Well done

$$ii) V = (3, 4) \quad (1)$$

$$iii) (x-3)^2 = 4 \cdot 2(y-4)$$

$$(x-3)^2 = 8(y-4) \quad (1)$$

e) False as there are 2 chances of having B or G. (1) Comm

Well done

Well done.

Probability tree helps.

$$5 a) \log_m 3 + \log_m (2 \times 3) + \log_m (2^2 \times 3) + \dots$$

$$\therefore \text{A.P. } a = \log_m 3, d = \log_m 2 \quad (1)$$

$$S_{10} = \frac{10}{2} (2 \times \log_m 3 + 9 \log_m 2)$$

$$= 5 (2 \times 0.48 + 9 \times 0.3)$$

$$= 18.3 \quad (1)$$

b) i) $\frac{dy}{dx}$ is +ve for all x

except when $x=1$ and it equals 0. This implies the curve

This is an AP not a GP
Use log laws to break down each term, then you will see the difference of $\log_m 2$

2 marks and 2 parts to the question so 2 statements must be made. 1 - comment on sign of $\frac{dy}{dx}$, 2 - what does this imply.

Reas
3

Reas
3

Calc
4

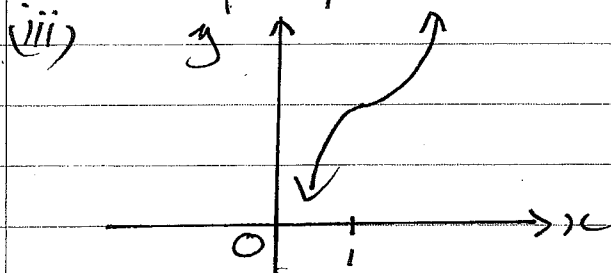
① is an increasing fn. except where $x=1$.

ii) When $x=1$, there is a stationary point on $y=f(x)$ as $\frac{dy}{dx}=0$

and as $x < 1$ $\frac{dy}{dx} > 0$

and as $x > 1$ $\frac{dy}{dx} > 0$

ie. at $x=1$ there is a horizontal point of inflexion ①



you must clearly show the horizontal POI at $x=1$

c) (i) $t=0 \therefore P = 50000$ ①

(ii) $2500 = 50000 e^{-0.08t}$

$$\textcircled{1} \frac{1}{2} = e^{-0.08t}$$

$$\ln \frac{1}{2} = -0.08t$$

$$t = \ln \frac{1}{2} \div -0.08$$

$$\approx 9 \text{ yrs} \textcircled{1}$$

(iii) $\frac{dP}{dt} = 50000 \times -0.08 e^{-0.08t}$ ①

when $t=5$, $= -2681.280$

ie. 2681 people/yr. ①

Be careful when transcribing numbers. A few used 5000 instead of 50000

6. a) i) $y = \log_e x$

$$x = e^y \textcircled{1}$$

$$A = \int_{\ln 8}^{\ln 8} e^y dy = [e^y]_{\ln 8}^{\ln 8} \textcircled{1}$$

$$= e^{\ln 8} - e^0 = 8 - 1 = 7 \textcircled{1}$$

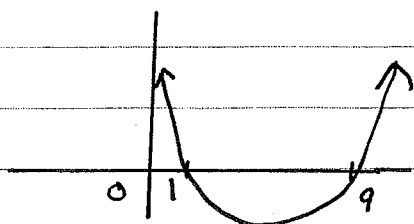
(ii) $\int_1^8 \ln x dx = 8 \times \ln 8 - 7$

Calc
4

$$= 8 \ln 8 - 7$$

(1)

b) i)



(1)

$$\therefore 1 < k < 9$$

ii) +ve definite $a > 0$ & $\Delta < 0$

ie. $k > 0$ and

$$(k+3)^2 - 4 \cdot k \cdot 4 < 0$$

(1)

$$k^2 - 10k + 9 < 0$$

$$(k-9)(k-1) < 0$$

$$\therefore 1 < k < 9$$

Comm
1

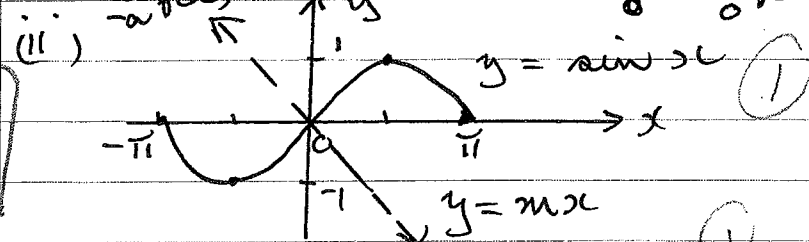
iii) +ve definite means the value of $kx^2 + (k+3)x + 4$ is always +ve $\therefore$ can't be -ve definite

Comm
2

c) i) sine is an odd function

$\therefore \int_{-a}^a f(x) dx = 0$ and $\int_{-a}^a f(x) dx$ is -ve & same size as $\int_0^a f(x) dx$

Reas
2



$\therefore$ only 1 possible soln.

$$7. a) S_3 = \frac{a(1-r^3)}{1-r} = 19$$

(1)

$$S_{\infty} = \frac{a}{1-r} = 27$$

(1)

(2)

$$\text{sub. (2) in (1)} \quad 27(1-r^3) = 19$$

$$1-r^3 = \frac{19}{27}$$

$$1 - \frac{19}{27} = r^3$$

$$\therefore r^3 = \frac{8}{27}$$

If your calculations get out of hand stop and look for another method of solution it is only worth 1 mark.

$$\therefore r = \frac{2}{3}$$

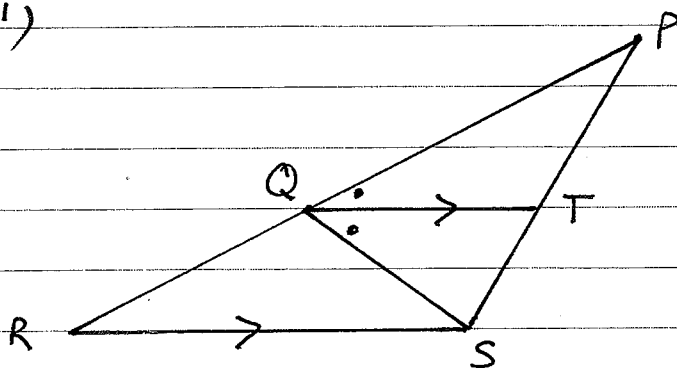
$$(ii) \frac{a}{1 - \frac{2}{3}} = 27$$

$$3a = 27$$

$$\therefore a = 9$$

$$(iii) T = 9 \times \left(\frac{2}{3}\right)^4 = \frac{16}{9}$$

b) (i)



Comm

(i) $\angle TQS = \angle QSR$ (alt $\angle$'s = on $\parallel$ lines)

(ii) $\angle QRS = \angle PQT$ (corr $\angle$'s = on $\parallel$ lines)

$\therefore \angle QRS = \angle QSR$ from above

$\therefore \triangle QRS$ is isosceles

(iii) $\frac{PQ}{QR} = \frac{PT}{TS}$ ($\parallel$ lines cut off = intercepts on transversals)

but $QR = QS$ as $\triangle QRS$ is isosceles

$$\therefore \frac{PT}{TS} = \frac{PQ}{QS}$$

$$c) \begin{array}{cc} \frac{I}{0.6P} & \frac{P}{0.4P} \\ & \frac{0.6F}{0.4F} \end{array}$$

$$(i) P(P) = 0.6 \times 0.4 = 0.24 \text{ or } 24\%$$

$$(ii) P(P \text{ one}) = 0.6 \times 0.6 = 0.36 \text{ or } 36\%$$

If your $r > 1$
then no CFFPE
for parts (ii) and
(iii) as can't find
 S_{∞}

Again make sure your
reasons are clear and
accurate. Take care
when writing out
angles i.e. don't mix
up $\angle QSR$ & $\angle QRS$.

Reas

2

Very well done!
The people
who drew tree diagrams
had the greatest
success

$$8. a) i) x = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\left(\frac{5\pi}{6}, \frac{1}{2}\right)$$

Use symmetry to find x .
Many left out $\frac{1}{2}$

Calc
12

$$\begin{aligned}
 \text{ii) } A &= \int_{\pi/6}^{\pi/2} \sin x - \cos 2x \, dx \\
 &= \left[-\cos x - \frac{1}{2} \sin 2x \right]_{\pi/6}^{\pi/2} \\
 &= \left(0 - \frac{1}{2} \times 0 \right) - \left(-\frac{\sqrt{3}}{2} - \frac{1}{2} \times \frac{\sqrt{3}}{2} \right) \\
 &= \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{4} = \frac{3\sqrt{3}}{4}
 \end{aligned}$$

Well done

Be careful when integrating

$$\begin{aligned}
 \text{b) i) } f(-x) &= \frac{1}{2}(e^{-x} + e^x) = f(x) \\
 \therefore \text{ even fn}
 \end{aligned}$$

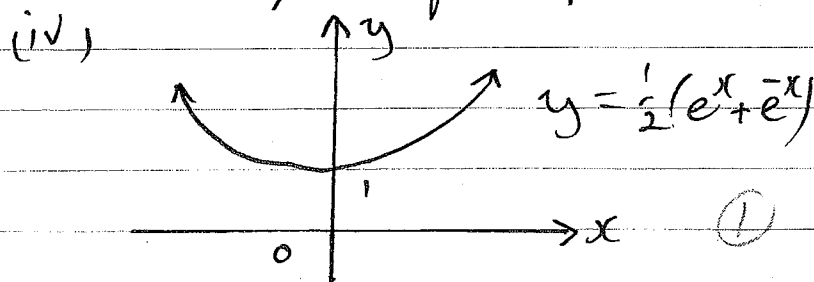
Well done.

$$\begin{aligned}
 \text{ii) } f'(x) &= \frac{1}{2}(e^x - e^{-x}) \\
 \frac{1}{2}(e^x - e^{-x}) &= 0 \\
 e^x &= e^{-x}
 \end{aligned}$$

Know your derivatives.

If solution is not obvious take logs of both sides.

$$\begin{aligned}
 \therefore x &= 0 \\
 f''(x) &= \frac{1}{2}(e^x + e^{-x}) \\
 f''(0) &= \frac{1}{2}(1+1) = 1 \\
 \therefore \text{ min at } (0, 1) \\
 \text{iii) } f''(x) &= \frac{1}{2}(e^x + e^{-x}) = 0 \\
 e^x &= -e^{-x} \\
 \therefore \text{ no soln.} \\
 \therefore \text{ no pt. of inflex}
 \end{aligned}$$



Put y intercept on graph

$$9 \text{ a) i) } A_1 = 50000 \times 1.08 - M$$

$$\text{ii) } A_2 = 50000 \times 1.08^2 - M(1.08 + 1)$$

$$\begin{aligned}
 &\vdots \\
 A_{20} &= 50000 \times 1.08^{20} - M(1.08^{19} + \dots + 1)
 \end{aligned}$$

$$\begin{aligned}
 \text{iii) } M &= \frac{50000 \times 1.08^{20}}{1.08^{19} + \dots + 1} \\
 &= \frac{50000 \times 1.08^{20} (1.08 - 1)}{1 (1.08^{20} - 1)}
 \end{aligned}$$

Many were confused with indices inside brackets

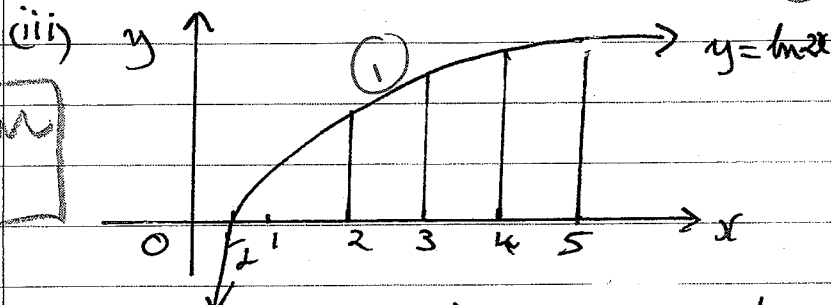
$$M = \$5092.60$$

b)(i)	x	$\ln 2x$	w	$w \times \ln 2x$
	2	1.386	1	1.386
(2)	3	1.792	2	3.584
	4	2.079	2	4.159
	5	2.303	1	2.303

Read the question.
You were asked to
complete the table
correct to 3 d.p.

$$11.432$$

$$(ii) A = \frac{1}{2} \times 11.432 = 5.72$$



To get the mark for
the graph you
needed to show the
correct x -intercept

Since the curve is concave down,
all the trapeziums will lie
under the curve $\therefore$ the approx
will be less than the exact
value of the integral

$$(iv) \frac{d}{dx} (x \ln 2x - x)$$

$$= \ln 2x + \frac{2}{2x} \cdot x - 1$$

$$= \ln 2x + 1 - 1 = \ln 2x$$

$$(v) \int_2^5 \ln 2x \, dx = [x \ln 2x - x]$$

$$= (5 \ln 10 - 5) - (2 \ln 4 - 2)$$

$$= \ln 100000 - \ln 16 - 3$$

$$= \frac{\ln 100000}{16} = \ln 6250 - 3$$

Many tried to
fudge this answer
you should show
clearly especially
 $\frac{2}{2x} \cdot x$

Careful how you
simplify this answer

$$10. a) P(nD) = 95\%$$

$$(ii) \begin{array}{l} 0.05 \overline{) D} \\ 0.95 \overline{) nD} \end{array} \quad \begin{array}{l} 0.05 \overline{) D} \\ 0.95 \overline{) nD} \end{array} \quad \text{etc}$$

$$(0.95)^n > 0.5 \quad (1)$$

$$\ln(0.95^n) > \ln 0.5$$

$$n \ln 0.95 > \ln 0.5$$

$$n < \frac{\ln 0.5}{\ln 0.95}$$

$$n < 13.51340733$$

$$\therefore n = 13 \quad (1)$$

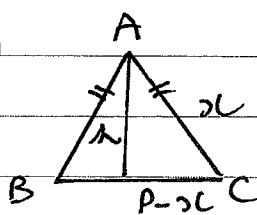
$$b) V = \int_0^1 \frac{1}{2x+1} dx \quad (1)$$

$$= \frac{\pi}{2} [\ln(2x+1)]_0^1 \quad (1)$$

$$= \frac{\pi}{2} (\ln 3 - \ln 1)$$

$$= \frac{\pi}{2} \ln 3 \quad (1)$$

c) (i)



$$h^2 = x^2 - (P-x)^2$$

$$= x^2 - (P^2 - 2Px + x^2)$$

$$= -P^2 + 2Px \quad (1)$$

$$\therefore h = \sqrt{2Px - P^2}$$

$$\therefore A = \frac{1}{2} (2P - 2x) \sqrt{2Px - P^2} \quad (1)$$

$$= (P-x) \sqrt{2Px - P^2}$$

$$(ii) \frac{dA}{dx} = -1 \sqrt{2Px - P^2} + (P-x) \frac{1}{2} 2P (2Px - P^2)^{-1/2} \quad (1)$$

$$= \frac{P(P-x)}{\sqrt{2Px - P^2}} - \sqrt{2Px - P^2}$$

$$(iii) \sqrt{2Px - P^2} = \frac{P(P-x)}{\sqrt{2Px - P^2}}$$

$$2Px - P^2 = P^2 - Px$$

$$3Px = 2P^2$$

$$x = \frac{2P}{3} \quad (1)$$

$$\therefore \text{sides are } \frac{2P}{3}, \frac{2P}{3}, 2P - \frac{4P}{3} = \frac{2P}{3}$$

$$\therefore \text{perimeter} = 3 \times \frac{2P}{3} = 2P$$

x	$\frac{P}{3}$	$\frac{2P}{3}$	P
$\frac{dA}{dx}$	+ve	0	-ve

$$\therefore \text{max area when } x = \frac{2P}{3} \quad (1)$$